EE 66
Fall 2026

Signals, Dynamics, and Information

Homework 7

This homework is due Friday, October 16, 2026, at 23:59.

Submission Format

Your homework submission should consist of a single PDF file that contains all of your answers (any hand-written answers should be scanned) as well as your IPython notebook saved as a PDF.

If you do not attach a PDF "printout" of your IPython notebook, you will not receive credit for problems that involve coding. Make sure that your results and your plots are visible.

On occasion, a problem set contains one more problems designated as optional. Your solutions to such problems are not to be evaluated, and will not count toward your grade. Nevertheless, you are still responsible for learning the subject matter within their scope. In other words, we may test you on their scope.

1. Least squares and QR factorization (Adapted from VMLS 12.8)

A quick review of the QR factorization. Writing $\mathbf{A} = \mathbf{QR}$ splits $\mathbf{A}$ into a matrix $\mathbf{Q}$ whose columns are orthonormal, meaning $\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}$, together with an upper triangular $\mathbf{R}$ whose entries are the coefficients that rebuild $\mathbf{a}_i$ (these are columns of $\mathbf{A}$) out of $\mathbf{q}_1, \ldots, \mathbf{q}_i$ and none of the later ones. Justify this by writing out an arbitrary orthonormal $\mathbf{Q}$ and upper-triangular $\mathbf{R}$, and remind yourself of the linear-combinations-of-the-columns view of matrix multiplication!

Since each $\mathbf{a}_i$ can be built from the $\mathbf{q}_j$, and each $\mathbf{q}_j$ can be built back out of the $\mathbf{a}_i$,

$$\mathrm{Col}(\mathbf{Q}) = \mathrm{Col}(\mathbf{A}).$$

You will not need to carry out a QR factorization anywhere in this problem. Every part below follows from the two facts just stated, namely $\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}$ and $\mathrm{Col}(\mathbf{Q}) = \mathrm{Col}(\mathbf{A})$, so it is fine if the procedure for computing $\mathbf{Q}$ and $\mathbf{R}$ is not fresh in your memory.

Suppose $\mathbf{A}$ is an $m \times n$ matrix with linearly independent columns and QR factorization $\mathbf{A} = \mathbf{QR}$, and $\mathbf{b} \in \mathbb{R}^m$. The vector $\mathbf{A}\hat{\mathbf{x}}$ is the linear combination of the columns of $\mathbf{A}$ that is closest to the vector $\mathbf{b}$, i.e., it is the projection of $\mathbf{b}$ onto the column space of $\mathbf{A}$.

(a) Show that $\mathbf{A}\hat{\mathbf{x}} = \mathbf{QQ}^\top\mathbf{b}$, i.e. that $\mathbf{QQ}^\top\mathbf{b}$ is the projection of $\mathbf{b}$ onto the column space of $\mathbf{A}$.
Hint: how do you project onto an orthonormal basis?

Solution: We recall that for the QR decomposition $\mathbf{A} = \mathbf{QR}$, the column space of $\mathbf{Q}$ is actually identical to the column space of $\mathbf{A}$. As a result, if we show that $\mathbf{QQ}^\top\mathbf{b}$ is a projection of $\mathbf{b}$ onto the column space of $\mathbf{Q}$, then it must also be a projection onto the column space of $\mathbf{A}$.

Because the columns of $\mathbf{Q}$ form an orthonormal basis, we know that a projection onto their span is identical to taking a linear combination (in this case the sum) of the vector's projection onto each basis vector individually. In other words:

$$\mathbf{A}\hat{\mathbf{x}} = \mathrm{proj}_{\mathrm{Col}(\mathbf{Q})}\mathbf{b} = \mathrm{proj}_{\mathbf{q}_1}\mathbf{b} + \mathrm{proj}_{\mathbf{q}_2}\mathbf{b} + \cdots + \mathrm{proj}_{\mathbf{q}_n}\mathbf{b}$$

Observe via the projection formula that:

$$\mathrm{proj}_{\mathbf{q}_i}\mathbf{b} = \frac{\langle \mathbf{q}_i, \mathbf{b}\rangle}{\|\mathbf{q}_i\|^2}\mathbf{q}_i = \langle \mathbf{q}_i, \mathbf{b}\rangle \mathbf{q}_i$$

UCB EE 66, Fall 2026, Homework 7, All Rights Reserved. This may not be publicly shared without explicit permission.

Therefore,

$$\mathbf{A}\hat{\mathbf{x}} = \langle \mathbf{q}_1, \mathbf{b} \rangle \mathbf{q}_1 + \langle \mathbf{q}_2, \mathbf{b} \rangle \mathbf{q}_2 + \cdots + \langle \mathbf{q}_n, \mathbf{b} \rangle \mathbf{q}_n$$

At the same time, we can break down $\mathbf{Q}\mathbf{Q}^\top \mathbf{b}$ as follows:

$$\begin{aligned}
\mathbf{Q}\mathbf{Q}^\top \mathbf{b} &= \begin{bmatrix} | & | & \cdots & | \\ \mathbf{q}_1 & \mathbf{q}_2 & \cdots & \mathbf{q}_n \\ | & | & \cdots & | \end{bmatrix} \begin{bmatrix} - & \mathbf{q}_1^\top & - \\ - & \mathbf{q}_2^\top & - \\ \vdots & \vdots & \vdots \\ - & \mathbf{q}_n^\top & - \end{bmatrix} \mathbf{b} \\
&= \begin{bmatrix} | & | & \cdots & | \\ \mathbf{q}_1 & \mathbf{q}_2 & \cdots & \mathbf{q}_n \\ | & | & \cdots & | \end{bmatrix} \begin{bmatrix} \langle \mathbf{q}_1, \mathbf{b} \rangle \\ \langle \mathbf{q}_2, \mathbf{b} \rangle \\ \\ \langle \mathbf{q}_n, \mathbf{b} \rangle \end{bmatrix} \\
&= \langle \mathbf{q}_1, \mathbf{b} \rangle \mathbf{q}_1 + \langle \mathbf{q}_2, \mathbf{b} \rangle \mathbf{q}_2 + \cdots + \langle \mathbf{q}_n, \mathbf{b} \rangle \mathbf{q}_n
\end{aligned}$$

Therefore, we can see that

$$\mathbf{Q}\mathbf{Q}^\top \mathbf{b} = \langle \mathbf{q}_1, \mathbf{b} \rangle \mathbf{q}_1 + \langle \mathbf{q}_2, \mathbf{b} \rangle \mathbf{q}_2 + \cdots + \langle \mathbf{q}_n, \mathbf{b} \rangle \mathbf{q}_n = \mathbf{A}\hat{\mathbf{x}}$$

just as we desired to show.

(b) Show that $\|\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}\|^2 = \|\mathbf{b}\|^2 - \|\mathbf{Q}^\top \mathbf{b}\|^2$. (This is the square of the distance between $\mathbf{b}$ and the closest linear combination of the columns of $\mathbf{A}$.)

Solution: We can begin by expanding our norm expression as follows:

$$\begin{aligned}
\|\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}\|^2 &= \langle \mathbf{A}\hat{\mathbf{x}} - \mathbf{b}, \mathbf{A}\hat{\mathbf{x}} - \mathbf{b} \rangle \\
&= (\mathbf{A}\hat{\mathbf{x}} - \mathbf{b})^\top (\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}) \\
&= (\hat{\mathbf{x}}^\top \mathbf{A}^\top - \mathbf{b}^\top)(\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}) \\
&= \hat{\mathbf{x}}^\top \mathbf{A}^\top \mathbf{A}\hat{\mathbf{x}} - \hat{\mathbf{x}}^\top \mathbf{A}^\top \mathbf{b} - \mathbf{b}^\top \mathbf{A}\hat{\mathbf{x}} + \mathbf{b}^\top \mathbf{b}
\end{aligned}$$

From here, we can plug in the result from the previous subpart, that $\mathbf{A}\hat{\mathbf{x}} = \mathbf{Q}\mathbf{Q}^\top \mathbf{b}$:

$$\begin{aligned}
\|\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}\|^2 &= \hat{\mathbf{x}}^\top \mathbf{A}^\top \mathbf{A}\hat{\mathbf{x}} - \hat{\mathbf{x}}^\top \mathbf{A}^\top \mathbf{b} - \mathbf{b}^\top \mathbf{A}\hat{\mathbf{x}} + \mathbf{b}^\top \mathbf{b} \\
&= \mathbf{b}^\top \mathbf{Q}\mathbf{Q}^\top \mathbf{Q}\mathbf{Q}^\top \mathbf{b} - \mathbf{b}^\top \mathbf{Q}\mathbf{Q}^\top \mathbf{b} - \mathbf{b}^\top \mathbf{Q}\mathbf{Q}^\top \mathbf{b} + \mathbf{b}^\top \mathbf{b} \\
&= \mathbf{b}^\top \mathbf{Q}\mathbf{Q}^\top \mathbf{b} - 2\mathbf{b}^\top \mathbf{Q}\mathbf{Q}^\top \mathbf{b} + \|\mathbf{b}\|^2 \\
&= \|\mathbf{b}\|^2 - \mathbf{b}^\top \mathbf{Q}\mathbf{Q}^\top \mathbf{b} \\
\|\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}\|^2 &= \|\mathbf{b}\|^2 - \|\mathbf{Q}^\top \mathbf{b}\|^2
\end{aligned}$$

2. Linear Regression

An object moves along a straight line at an unknown but constant velocity $\dot{y}(t) = \beta$, where $\dot{y}(t) = dy(t)/dt$ denotes the derivative of y with respect to time t. We want to determine its velocity and initial position, which characterize the straight line

$$\forall t \in \mathbb{R} \qquad y(t) = \alpha + \beta t,$$

where α is the unknown position of the object at time $t = 0$—that is, $y(0) = \alpha$.

To do this, we measure the position of the object at M time instants $t_1, t_2, \ldots, t_M$. We denote our measurements by $y_1, \ldots, y_M$. Unfortunately, our measurements are prone to error, caused by imperfections of our measurement instruments. What we have is the following set of M equations in two unknowns:

$$\begin{aligned}
\alpha + t_1 \beta &= y_1 \\
&\vdots \\
\alpha + t_m \beta &= y_m \\
&\vdots \\
\alpha + t_M \beta &= y_M.
\end{aligned}$$

We can write our measurement equations in matrix-vector form $\mathbf{Ax} = \mathbf{y}$ as follows:

$$\underbrace{\begin{bmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_m \\ \vdots & \vdots \\ 1 & t_M \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} \alpha \\ \beta \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} y_1 \\ \vdots \\ y_m \\ \vdots \\ y_M \end{bmatrix}}_{\mathbf{y}} \tag{1}$$

We know the least-squares solution $\widehat{\mathbf{x}}$ to Equation (1) is given by

$$\widehat{\mathbf{x}} = \left(\mathbf{A}^\mathsf{T}\mathbf{A}\right)^{-1}\mathbf{A}^\mathsf{T}\mathbf{y}.$$

(a) Determine an expression for each of $\mathbf{A}^\mathsf{T}\mathbf{A}$, $\left(\mathbf{A}^\mathsf{T}\mathbf{A}\right)^{-1}$, and $\mathbf{A}^\mathsf{T}\mathbf{y}$. Note that these expressions wind up pretty gnarly; do your best to make them as simple as possible but you will still have some summations. Each of your expressions should be in terms of an appropriate subset of M, t_m, and y_m, where $m = 1, \ldots, M$.

Note: Do NOT carry out the multiplication

$$\left(\mathbf{A}^\mathsf{T}\mathbf{A}\right)^{-1}\mathbf{A}^\mathsf{T}\mathbf{y}.$$

Solution: $\mathbf{A}^\top\mathbf{A} = \begin{bmatrix} 1 & 1 & \ldots & 1 \\ t_1 & t_2 & \ldots & t_m \end{bmatrix} \begin{bmatrix} 1 & t_1 \\ 1 & t_2 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix} = \begin{bmatrix} \sum_{m=1}^{M} 1 & \sum_{m=1}^{M} 1 \cdot t_m \\ \sum_{m=1}^{M} t_m \cdot 1 & \sum_{m=1}^{M} t_m \cdot t_m \end{bmatrix} = \begin{bmatrix} M & \sum_{m=1}^{M} t_m \\ \sum_{m=1}^{M} t_m & \sum_{m=1}^{M} t_m^2 \end{bmatrix}$

Note that $\mathbf{A}^\top\mathbf{A}$ is symmetric as expected.

$$\mathbf{A}^\top \mathbf{y} = \begin{bmatrix} 1 & 1 & \dots & 1 \\ t_1 & t_2 & \dots & t_m \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix} = \begin{bmatrix} \sum_{m=1}^{M} y_m \\ \sum_{m=1}^{M} t_m y_m \end{bmatrix}$$

Using the inverse formula, we get that $(\mathbf{A}^\top \mathbf{A})^{-1} = \frac{1}{M \sum_{m=1}^{M} t_m^2 - (\sum_{m=1}^{M} t_m)^2} \begin{bmatrix} \sum_{m=1}^{M} t_m^2 & -\sum_{m=1}^{M} t_m \\ -\sum_{m=1}^{M} t_m & M \end{bmatrix}$

You should be relieved that we did not ask you to determine the least squares solution $\widehat{\mathbf{x}}$ itself, because the expression for it would not be pretty! In what follows, we'll apply the Gram-Schmidt method to orthogonalize (but not orthonormalize) the columns of $\mathbf{A} = [\mathbf{a}_1 \ \mathbf{a}_2]$. In particular, let the columns of $\mathbf{A}$ be given by

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} = \mathbf{1} \qquad \text{and} \qquad \mathbf{a}_2 = \begin{bmatrix} t_1 \\ \vdots \\ t_M \end{bmatrix},$$

where $\mathbf{1}$ denotes the all-ones vector of size M—that is, the vector each of whose M entries is 1. Take as your first orthogonal vector $\mathbf{z}_1 = \mathbf{a}_1$; note that $\mathbf{z}_1$ does not have unit length, and we won't bother normalizing its length.

A quick review of Gram-Schmidt. The word "orthogonal" puts a smile on the face of any student of linear algebra (remind yourself of why this is!) and so we often find it convenient to replace an arbitrary set of vectors with another *orthogonal* set spanning exactly the same space. Recall that two vectors fail to be orthogonal when they share a component in some common direction. It should feel reasonable therefore to strip away that shared part from one of the vectors, leaving only the component orthogonal to the other. Convince yourself that this won't change the plane spanned by the pair. Gram-Schmidt generalizes this to a set of vectors by repeating the process, one vector at a time: add the first vector to your orthogonal set S as it is, and iteratively add to S more vectors after subtracting off the component already spanned by S.

Back to the problem at hand: here there are only two vectors, so it takes a single step. After setting $\mathbf{z}_1 = \mathbf{a}_1$,

$$\mathbf{z}_2 = \mathbf{a}_2 - \frac{\langle \mathbf{a}_1, \mathbf{a}_2 \rangle}{\langle \mathbf{a}_1, \mathbf{a}_1 \rangle} \mathbf{a}_1,$$

where the subtracted term is the part of $\mathbf{a}_2$ already lying along $\mathbf{a}_1$; you can check that $\langle \mathbf{z}_1, \mathbf{z}_2 \rangle = 0$.

(b) Show that the second orthogonal vector $\mathbf{z}_2$ is given by

$$\mathbf{z}_2 = \big(\mathbf{I} - \mathbf{P}_1\big)\mathbf{a}_2, \tag{2}$$

where $\mathbf{I}$ is the identity matrix of appropriate size, and $\mathbf{P}_1$ is the matrix given by

$$\mathbf{P}_1 = \frac{\mathbf{a}_1 \mathbf{a}_1^\top}{\langle \mathbf{a}_1, \mathbf{a}_1 \rangle}.$$

You can do this by proving that $\mathbf{z}_2$, as given by Equation (2), is, in fact, orthogonal to $\mathbf{z}_1 = \mathbf{a}_1$.

Solution: The projection matrix onto span $\{a_1\}$ is $\mathbf{P}_1 = \frac{\mathbf{a}_1 \mathbf{a}_1^\top}{\mathbf{a}_1^\top \mathbf{a}_1}$.

The second orthogonal vector is $\mathbf{z}_2 = \mathbf{a}_2 - \mathbf{P}_1\mathbf{a}_2$. So we get that

$$\mathbf{z}_2 = \mathbf{a}_2 - \mathbf{P}_1\mathbf{a}_2 = (\mathbf{I} - \mathbf{P}_1)\mathbf{a}_2$$

as desired.

To check that they are orthogonal, we have that:

$$\begin{aligned}\langle \mathbf{z}_1, \mathbf{z}_2 \rangle &= \mathbf{a}_1^\top(\mathbf{a}_2 - \mathbf{P}_1\mathbf{a}_2) \\ &= \mathbf{a}_1^\top\mathbf{a}_2 - \mathbf{a}_1^\top \frac{\mathbf{a}_1\mathbf{a}_1^\top}{\mathbf{a}_1^\top\mathbf{a}_1}\mathbf{a}_2 \\ &= \mathbf{a}_1^\top\mathbf{a}_2 - \mathbf{a}_1^\top\mathbf{a}_2 \\ &= 0.\end{aligned}$$

In this case, the $\mathbf{P}_1$ matrix is the projection matrix onto span $\{a_1\}$, so $\mathbf{P}_1\mathbf{a}_2$ is the projection of $\mathbf{a}_2$ onto $\mathbf{a}_1$.

(c) Determine a reasonably simple form for $\mathbf{z}_2$. Your expressions for the entries in $\mathbf{z}_2$ should be in terms of M, t_m, and $\bar{t}$, where

$$\bar{t} = \frac{1}{M}\sum_{m=1}^{M} t_m.$$

Explain what the orthogonalization of $\mathbf{a}_1$ and $\mathbf{a}_2$ to the forms $\mathbf{z}_1 = \mathbf{a}_1$ and $\mathbf{z}_2 = \big(\mathbf{I} - \mathbf{P}_1\big)\mathbf{a}_2$ means in terms of relocating the origin of the time axis along which we've made our measurements.

Solution: We have that $\mathbf{z}_1 = \mathbf{1}$.

Using part (b), we have that $\mathbf{P}_1 = \frac{1}{M}\mathbf{1}\mathbf{1}^\top = \frac{1}{M}\underbrace{\begin{bmatrix} 1 & \dots & 1 \\ \vdots & & \vdots \\ 1 & \dots & 1 \end{bmatrix}}_{\text{all-ones matrix}}$.

The second orthogonal vector is

$$\begin{aligned}\mathbf{z}_2 &= (\mathbf{I} - \mathbf{P}_1)\mathbf{a}_2 \\ &= \begin{bmatrix} t_1 \\ \vdots \\ t_m \end{bmatrix} - \frac{1}{M}\begin{bmatrix} 1 & \dots & 1 \\ \vdots & & \vdots \\ 1 & \dots & 1 \end{bmatrix}\begin{bmatrix} t_1 \\ \vdots \\ t_m \end{bmatrix} \\ &= \begin{bmatrix} t_1 \\ \vdots \\ t_m \end{bmatrix} - \frac{\sum_{m=1}^{M} t_m}{M}\begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} t_1 - \bar{t} \\ \vdots \\ t_m - \bar{t} \end{bmatrix}.\end{aligned}$$

Interpretation: The second orthogonal vector $\mathbf{z}_2$ is obtained by subtracting the arithmetic mean of the measurement from each measurement time. That is, we shift the origin of time to $\bar{t}$. This corresponds to looking at $y(\tau) = \widehat{\alpha} + \widehat{\beta}\tau$, where $\tau = t - \bar{t}$, so $y(t) = (\widehat{\alpha} - \widehat{\beta}\bar{t}) + \widehat{\beta}t$. Therefore $\alpha = \widehat{\alpha} - \widehat{\beta}\bar{t}$ and $\beta = \widehat{\beta}$.

(d) Suppose we measure the position of the moving object at time instants $t_1 = 0$, $t_2 = 1$, and $t_3 = 2$ seconds, and that we've registered the following position values:

$$y(0) = 1, \qquad y(1) = 3, \qquad y(2) = 5.$$

Determine α and β, which define the straight line that best fits the data in a least-squares sense.

Solution: First, given the 3 measurement times, we figure out $\bar{t} = \frac{t_1+t_2+t_3}{3} = 1$.

Let $\tau = t - \bar{t} = t - 1$, so $y(\tau) = \widehat{\alpha} + \widehat{\beta}\,\tau$ where $\tau = -1, 0, 1$.

For the least squares expression $\widehat{\mathbf{A}}\widehat{\mathbf{x}} \approx \mathbf{y}$:

$$\widehat{\mathbf{A}} = \begin{bmatrix} 1 & \tau_1 \\ 1 & \tau_2 \\ 1 & \tau_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}.$$

Solve for $\widehat{\mathbf{x}}$:

$$\mathbf{A}^\top \mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix},$$

so

$$(\widehat{\mathbf{A}}^\top \widehat{\mathbf{A}})^{-1} = \begin{bmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}.$$

$$\widehat{\mathbf{A}}^\top \widehat{\mathbf{y}} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 9 \\ 4 \end{bmatrix}.$$

$$\widehat{\mathbf{x}} = (\widehat{\mathbf{A}}^\top \widehat{\mathbf{A}})^{-1} \widehat{\mathbf{A}}^\top \mathbf{y} = \begin{bmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 9 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

From part (c), $\beta = \widehat{\beta} = 2$ and $\alpha = \widehat{\alpha} - \widehat{\beta}\bar{t} = 3 - 1 \cdot 2 = 1$, so $y(t) = 1 + 2t$.

3. Braincells

After making your way through the last homework assignment, your remaining braincells are in disarray. Let's work together to get them back in order!

For all subparts of this problem, let $\mathbf{B} \in \mathbb{R}^{n \times n}$ be an *adjacency matrix* representing the connections between each of your braincells with the other braincells. In particular, entry $\mathbf{B}_{ij}$ represents the strength of communication from braincell j to braincell i, specifically $j \to i$, where $\mathbf{B}_{ij} = 0$ if there is no communication between braincell j and braincell i at all. Note that braincells do not communicate with themselves, so all diagonal entries should be 0.

Here are common types of matrices and what they look like:

Symmetric:	$\mathbf{A} = \mathbf{A}^\top$, or equivalently $A_{ij} = A_{ji}$.
Upper triangular:	$A_{ij} = 0$ whenever $i > j$.
Lower triangular:	$A_{ij} = 0$ whenever $i < j$.
Diagonal:	$A_{ij} = 0$ whenever $i \neq j$.
Zero matrix:	$\mathbf{A} = \mathbf{0}$, so every entry is 0.
Identity matrix:	$\mathbf{A} = \mathbf{I}$, with 1s on the diagonal and 0s elsewhere.

(a) At the very beginning, all of your braincells are disconnected from one another. Denote this adjacency matrix as $\mathbf{A}$. What kind of matrix is $\mathbf{A}$?

Solution:

Since all of our braincells are disconnected from one another, we can see that $\mathbf{A}$ will be a matrix of all zeros. As a result, $\mathbf{A}$ will be the zero matrix, $\mathbf{0}_{n \times n}$. Note that the zero matrix is also symmetric, upper triangular, lower triangular, and diagonal.

(b) After a good night's rest, all of your brain connections are beginning to come back. In particular, the strength of communication between every pair of braincells has increased by 1. Assuming that the adjacency matrix before going to sleep was represented by $\mathbf{B}$, express this new adjacency matrix, $\mathbf{B}'$, in terms of $\mathbf{B}$ and constant matrices, as well as any scalar multiples.

Note: Recall in the problem statement that braincells do not communicate with themselves. Your new $\mathbf{B}'$ *should preserve this property.*

Solution:

Per the problem description, the strength of communication between every pair of braincells has increased by 1. Therefore, we would like to increment every entry of $\mathbf{B}$ by 1. This would be equivalent to the following:

$$\mathbf{B}'_{ij} = \mathbf{B}_{ij} + 1, \ \forall\, i, j \in \{1, 2, \ldots, n\}$$

However, as the note reminds us, all of our diagonal entries should remain 0, since we assume braincells do not communicate with themselves. Therefore, we can rewrite our expression as follows:

$$\mathbf{B}'_{ij} = \begin{cases} \mathbf{B}_{ij} + 1 & \text{if } i \neq j, \\ \mathbf{B}_{ij} & \text{if } i = j, \end{cases} = \begin{cases} \mathbf{B}_{ij} + 1 & \text{if } i \neq j, \\ \mathbf{B}_{ij} + 1 - 1 & \text{if } i = j, \end{cases}$$

We can write this relationship in matrix form as:

$$\mathbf{B}' = \mathbf{B} + \mathbf{1}_{n \times n} - \mathbf{I}_n = \mathbf{B} + (\mathbf{1}_{n \times n} - \mathbf{I}_n)$$

(c) After consuming your favorite source of pre-midterm caffeine, the strength of communication between all of your braincells temporarily quadruples. Express this new adjacency matrix, $\mathbf{C}$, in terms of the adjacency matrix before caffeine, $\mathbf{B}$.

Solution:

Per the problem description, the strength of communication between every pair of braincells has quadrupled. Therefore, we would like to scale every entry of $\mathbf{B}$ by 4. This is equivalent to the following:

$\mathbf{C}_{ij} = 4\mathbf{B}_{ij}, \ \forall\, i, j \in \{1, 2, \ldots, n\}$

Written in matrix form, this means that:

$\mathbf{C} = 4\mathbf{B}$

(d) All ready to go, you attend your favorite TA's discussion. After collaborating with your peers on the new course content, the strength of communication from braincell k to braincell l increases by 10. The rest of the connections stay the same. Express the new adjacency matrix, $\mathbf{D}$, in terms of the adjacency matrix before discussion, $\mathbf{B}$, and a constant matrix, as well as any scalar multiples.

Note: If you cannot provide a closed-form mathematical expression for $\mathbf{D}$*, a qualitative expression is also acceptable.*

Solution:

In this case, the value of particular braincell connection is being incremented, and the rest stay the same. As a result, we can see that this is equivalent to:

$\mathbf{D} = \mathbf{B} + \mathbf{E}$, where $\mathbf{E}_{lk} = 10$ and all other entries of $\mathbf{E}$ are zero. In other words, we just increase the entry $\mathbf{B}_{lk}$ by 10 and the rest stay the same. Note that it is $\mathbf{B}_{lk}$ and not $\mathbf{B}_{kl}$ since the connection goes from $k \to l$.

(e) For this subpart, imagine that the strength of communication between braincells i and j is equal in both directions. In other words, consider the case where $\mathbf{B}_{ij} = \mathbf{B}_{ji}\ \forall\, i, j \in \{1, 2, \ldots, n\}$. In this case, what kind of matrix is $\mathbf{B}$?

Solution:

In this case, $\mathbf{B}$ is a symmetric matrix. Indeed, for every i, j,

$$(\mathbf{B}^\top)_{ij} = \mathbf{B}_{ji} = \mathbf{B}_{ij}.$$

Therefore, $\mathbf{B}^\top = \mathbf{B}$.

(f) As you're leaving discussion, you see Oski in the distance! Suppose neuron i receives information from your eyes and sends it to neuron v in your visual cortex. There is no direct connection from i to v; instead, the information must pass through neuron g in a part of your brain called the LGN (Lateral Geniculate Nucleus). Assume that communication strengths multiply along a route. What is the strength of communication from i to v along the route $i \to g \to v$?

Solution:

The connection from i to g has strength $\mathbf{B}_{gi}$, and the connection from g to v has strength $\mathbf{B}_{vg}$. Since strengths multiply along a route, the strength from i to v along this route is $\mathbf{B}_{vg}\mathbf{B}_{gi}$.

(g) Now, suppose information can pass from neuron i to neuron v through any one intermediate neuron. What is the total two-step communication strength from i to v along all such routes?

Solution:

For each possible intermediate neuron k, the strength along the route $i \to k \to v$ is $\mathbf{B}_{vk}\mathbf{B}_{ki}$. Adding the

contributions from every possible intermediate neuron gives

$$\sum_{k=1}^{n} \mathbf{B}_{vk}\mathbf{B}_{ki}.$$

(h) Use the definition of matrix multiplication to show that the expression from the previous subpart is the (v,i) entry of $\mathbf{B}^2$. Interpret $(\mathbf{B}^2)_{vi}$ in the context of braincell communication.

Solution:

By the definition of matrix multiplication,

$$(\mathbf{B}^2)_{vi} = \sum_{k=1}^{n} \mathbf{B}_{vk}\mathbf{B}_{ki}.$$

Thus, $(\mathbf{B}^2)_{vi}$ is the total strength of communication from braincell i to braincell v along all routes with exactly two connections.

(i) Suppose every direct communication strength is either 0 or 1: an entry is 1 exactly when the corresponding direct connection exists, and 0 otherwise. What does $(\mathbf{B}^2)_{ij}$ represent in this case? More generally, for a positive integer r, what does $(\mathbf{B}^r)_{ij}$ represent?

Solution:

When the entries of $\mathbf{B}$ are 0 or 1, each product $\mathbf{B}_{ik}\mathbf{B}_{kj}$ equals 1 exactly when both connections $j \to k$ and $k \to i$ exist. Thus, $(\mathbf{B}^2)_{ij}$ is the number of two-step walks from braincell j to braincell i.

More generally, $(\mathbf{B}^r)_{ij}$ is the number of walks of length r from braincell j to braincell i. Therefore, powers of an adjacency matrix can be used to count communication routes with any fixed number of steps.

(j) **Optional:** Suppose $\mathbf{B}$ can now have positive and negative entries, where positive entries of $\mathbf{B}$ represent excitatory connections and negative entries represent inhibitory connections. (In this context, excitatory means one neuron increases the activity of another, while inhibitory means one neuron decreases the other's activity). Suppose we know neuron i inhibits neuron g, and neuron g inhibits neuron v. What is the sign of the effect of i on v along the route $i \to g \to v$? Explain your answer in the context of braincell communication.

Solution:

Since i inhibits g, we have $\mathbf{B}_{gi} < 0$. Since g inhibits v, we have $\mathbf{B}_{vg} < 0$. Therefore,

$$\mathbf{B}_{vg}\mathbf{B}_{gi} > 0.$$

Thus, i has a positive indirect effect on v: inhibiting an inhibitory neuron reduces that neuron's inhibition of v. This is called *disinhibition*.

4. Study Spaces: Moffitt Library vs. Gateway

A campus is tracking the number of students who choose to study at each location each evening. At each timestep n, every student studies at one of two locations:

- Moffitt Library ($x_{\mathrm{M}}[n]$),
- Gateway ($x_{\mathrm{G}}[n]$).

We define the state vector:

$$\mathbf{x}[n] = \begin{bmatrix} x_{\mathrm{M}}[n] \\ x_{\mathrm{G}}[n] \end{bmatrix}.$$

Observed switching behavior:

- 70% of students at Moffitt return to Moffitt next time; 30% switch to Gateway.
- 60% of students at Gateway return to Gateway next time; 40% switch to Moffitt.

Thus,

$$\mathbf{x}[n+1] = A\mathbf{x}[n] + \mathbf{u}[n], \qquad A = \begin{bmatrix} 0.7 & 0.4 \\ 0.3 & 0.6 \end{bmatrix}.$$

External input (renovations). Moffitt Library has just completed major renovations, drawing additional new students beyond the usual returners. Each timestep during the promotion:

$$\mathbf{u}[n] = \begin{cases} \begin{bmatrix} 50 \\ 20 \end{bmatrix}, & n = 0, 1, \\ \mathbf{0}, & n \geq 2, \end{cases}$$

where the 50 represents new students directly attracted to Moffitt because of the renovations, while the 20 represents extra students who also choose Gateway because overall study activity on campus increases.

The initial state is $\mathbf{x}[0] = \begin{bmatrix} 100 \\ 50 \end{bmatrix}$.

(a) Compute $\mathbf{x}[1], \mathbf{x}[2], \mathbf{x}[3]$.

Solution: **Step** $0 \to 1$:

$$A\mathbf{x}[0] = \begin{bmatrix} 0.7 & 0.4 \\ 0.3 & 0.6 \end{bmatrix} \begin{bmatrix} 100 \\ 50 \end{bmatrix} = \begin{bmatrix} 90 \\ 60 \end{bmatrix}, \quad \mathbf{x}[1] = \begin{bmatrix} 90 \\ 60 \end{bmatrix} + \begin{bmatrix} 50 \\ 20 \end{bmatrix} = \begin{bmatrix} 140 \\ 80 \end{bmatrix}.$$

Step $1 \to 2$:

$$A\mathbf{x}[1] = \begin{bmatrix} 0.7 & 0.4 \\ 0.3 & 0.6 \end{bmatrix} \begin{bmatrix} 140 \\ 80 \end{bmatrix} = \begin{bmatrix} 130 \\ 90 \end{bmatrix}, \quad \mathbf{x}[2] = \begin{bmatrix} 130 \\ 90 \end{bmatrix} + \begin{bmatrix} 50 \\ 20 \end{bmatrix} = \begin{bmatrix} 180 \\ 110 \end{bmatrix}.$$

Step $2 \to 3$ (no input):

$$\mathbf{x}[3] = A\mathbf{x}[2] = \begin{bmatrix} 0.7 & 0.4 \\ 0.3 & 0.6 \end{bmatrix} \begin{bmatrix} 180 \\ 110 \end{bmatrix} = \begin{bmatrix} 170 \\ 120 \end{bmatrix}.$$

(b) Let the total number of students be $T[n] = x_{\mathrm{M}}[n] + x_{\mathrm{G}}[n] = \mathbf{1}^\top \mathbf{x}[n]$ with $\mathbf{1} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Show that

$$T[n+1] = T[n] + \mathbf{1}^\top \mathbf{u}[n],$$

and interpret this result.

Hint: Do the columns of A sum to 1?

Solution: Each column of A sums to 1, so $\mathbf{1}^\top A = \mathbf{1}^\top$. Then

$$\begin{aligned} T[n+1] &= \mathbf{1}^\top \mathbf{x}[n+1] = \mathbf{1}^\top (A\mathbf{x}[n] + \mathbf{u}[n]) \\ &= \mathbf{1}^\top \mathbf{x}[n] + \mathbf{1}^\top \mathbf{u}[n] = T[n] + \mathbf{1}^\top \mathbf{u}[n]. \end{aligned}$$

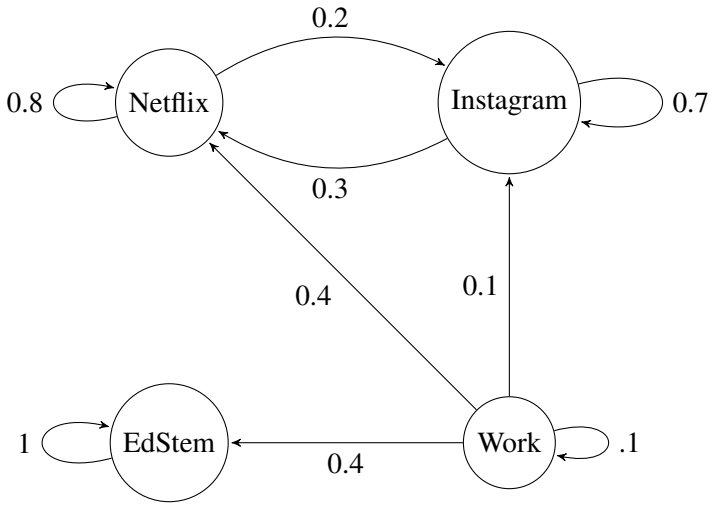
Interpretation: without renovations ($\mathbf{u}[n] = 0$), the total number of students is conserved—they only switch locations. Renovations add directly to the total.

(c) Suppose the renovations continue indefinitely with $\mathbf{u}[n] = \begin{bmatrix} 50 \\ 20 \end{bmatrix}$ $\forall n$. Without eigenvalues, describe qualitatively what happens as $n \to \infty$.

Solution: Each step adds $50 + 20 = 70$ new students, so $T[n]$ increases by exactly 70 each time. The matrix A redistributes students between Moffitt and Gateway. Over time, the proportions at the two locations approach the stable ratio $4 : 3$ (Moffitt to Gateway), but the total number of students keeps growing without limit because new people keep arriving every step.

5. Social Media

As a tech-savvy Berkeley student, the distractions of that little glowing box in your pocket is always calling you away from productive stuff like homework for your classes. You're curious—are you the only one who spends hours switching between Netflix and scrolling Instagram? How do you manage to pull yourself away from EdStem after posting an interesting question or expressing passionately your love of EE66? How do other students manage to get stuff done and balance staying up to date with the latest TV shows while maintaining their social media presence? Why is EdStem so addicting? You conduct an experiment, collect some data, and notice Berkeley students tend to follow a pattern of behavior similar to the figure below. So, for example, if $x = 100$ students are on Netflix, in the next timestep, 20 (i.e., $0.2 \cdot x$) of them will click on a link and move to Instagram, and 80 (i.e, $0.8 \cdot x$) will remain on Netflix.



This problem has an IPython component. Please attach a pdf printout of your IPython code to the HW in order to receive credit for problems that involve coding as stated in the submission format above.

(a) Let us define $x_N[n]$ as the number of students on Netflix at time-step n; $x_I[n]$ as the number of students on Instagram at time-step n; $x_E[n]$ as the number of students on EdStem at time-step n; and $x_W[n]$ as the number of students working at time-step n.

Let the state vector be: $\mathbf{x}[n] = \begin{bmatrix} x_N[n] & x_I[n] & x_E[n] & x_W[n] \end{bmatrix}^\top$. Derive the corresponding transition matrix $\mathbf{A}$.

Hint: A transition matrix, $\mathbf{A}$, is the matrix that transitions $\mathbf{x}[n]$, the vector at time-step n to $\mathbf{x}[n+1]$, the vector at time-step $n+1$. In other words: $\mathbf{x}[n+1] = \mathbf{A}\,\mathbf{x}[n]$.

Solution: Let us explicitly write the transition equation for each state and then we can use these to identify the state transition matrix.

$$
\begin{aligned}
x_N[n+1] &= 0.8 \cdot x_N[n] + 0.3 \cdot x_I[n] + 0.4 \cdot x_W[n] \\
x_I[n+1] &= 0.2 \cdot x_N[n] + 0.7 \cdot x_I[n] + 0.1 \cdot x_W[n] \\
x_E[n+1] &= 1 \cdot x_E[n] + 0.4 \cdot x_W[n] \\
x_W[n+1] &= 0.1 \cdot x_W[n]
\end{aligned}
$$

If $\mathbf{x}[n] = \begin{bmatrix} x_N[n] & x_I[n] & x_E[n] & x_W[n] \end{bmatrix}^\top$, then for $\mathbf{x}[n+1] = \mathbf{A}\,\mathbf{x}[n]$ we can identify the state transition matrix $\mathbf{A}$ as

$$\begin{bmatrix} 0.8 & 0.3 & 0 & 0.4 \\ 0.2 & 0.7 & 0 & 0.1 \\ 0 & 0 & 1 & 0.4 \\ 0 & 0 & 0 & 0.1 \end{bmatrix}$$

(b) There are 320 of you in the class. Suppose on a given Friday evening (the day when HW is due), there are 110 EE66 students on Netflix, 60 on Instagram, 10 on EdStem, and 140 actually doing work. In the next timestep, how many people will be doing each activity? In other words, after you apply the matrix once to reach the next timestep, what is the state vector?

Solution:

In order to calculate the state vector at the next timestep, we can use the equation $\mathbf{x}[n+1] = \mathbf{A}\,\mathbf{x}[n]$. Substituting the values for A and $\mathbf{x}[n]$, we get the following:

$$\begin{bmatrix} 0.8 & 0.3 & 0 & 0.4 \\ 0.2 & 0.7 & 0 & 0.1 \\ 0 & 0 & 1 & 0.4 \\ 0 & 0 & 0 & 0.1 \end{bmatrix} \begin{bmatrix} 110 \\ 60 \\ 10 \\ 140 \end{bmatrix} = \begin{bmatrix} 162 \\ 78 \\ 66 \\ 14 \end{bmatrix}$$

(c) Compute the sum of each column in the state transition matrix. What is the interpretation of this?

Solution:

Since each column's sum is equal to 1, the system is conservative. This means that we aren't losing (or gaining) students after each time step and the total number of students remains constant.

(d) Let $\mathbf{A}$ be the transition matrix you derived above. Complete your implementation in the provided Jupyter notebook.

i. Implement a simulator that takes $\mathbf{A}$ and an initial state $\mathbf{x}[0]$, and iterates $\mathbf{x}[n+1] = \mathbf{A}\,\mathbf{x}[n]$ until either $\|\mathbf{x}[n+1] - \mathbf{x}[n]\|_1 < 10^{-1}$ or $n = 1000$. Return all iterates $\mathbf{x}[0], \ldots, \mathbf{x}[T^*]$ where T^* is the last index. For this and all parts after, **do not** round $\mathbf{x}$ elements to the nearest whole number.

ii. Using your simulator with the initial state from part (a), $\mathbf{x}[0] = [110, 60, 10, 140]^\top$:

A. Plot $x_N[n], x_I[n], x_E[n], x_W[n]$ versus n on one figure with labeled axes and a legend.

B. For each iterate, compute $S[n] = \mathbf{1}^\top \mathbf{x}[n]$ and verify that $S[n] = 320$ for all n.

C. What is the steady-state vector $\widehat{\mathbf{x}}$ you observe at termination and what are the proportions of students in each state?

Solution: The trajectories should show $x_W[n]$ decreasing toward 0 while the streaming states level out to about 149 (46.4%) students on Netflix, 99 (31.0%) students on Instagram, 72 (22.6%) students on EdStem, and 0 students working. The steady-state vector $\widehat{\mathbf{x}}$ has all students on the streaming services and sums to 320. Also, $S[n]$ should stay constant at 320 for all n. See the solution notebook for an example of a correct implementation and plots.

(e) In the week before an exam, students realize that it's time to lock in, so everyone downloads a nifty new app that makes the user wait before they can start watching a streaming service. This does not, however, encourage users to turn *off* the streaming service.

Therefore, we can model this by modifying the Work column of the transition matrix to increase the liklihood that a working student keeps working. Hence we introduce a parameter α to represent the additional probability of staying in Work after the app is downloaded.

The original Work column of $\mathbf{A}$ is $\mathbf{w}^{(0)} = \begin{bmatrix}0.4 & 0.1 & 0.4 & 0.1\end{bmatrix}^\top$. For $\alpha \in [0, 0.9]$, define the modified Work column $\mathbf{w}(\alpha) : [0, 0.9] \to \mathbb{R}^4$ as

$$\mathbf{w}(\alpha) = \begin{bmatrix}0.4 - \alpha\frac{0.4}{0.9} & 0.1 - \alpha\frac{0.1}{0.9} & 0.4 - \alpha\frac{0.4}{0.9} & 0.1 + \alpha\end{bmatrix}^\top,$$

and construct $\mathbf{A}(\alpha) : [0, 0.9] \to \mathbb{R}^{4\times 4}$ by replacing only the Work column of $\mathbf{A}$ with $\mathbf{w}(\alpha)$ (all other columns unchanged). Convince yourself that this is a valid mathematical representation of the newly-downloaded app.

i. For $\alpha \in \{0.1, 0.2, \ldots, 0.9\}$, simulate $\mathbf{x}[n+1] = \mathbf{A}(\alpha)\mathbf{x}[n]$ to convergence and plot $x_W[n]$ versus n for each α (with labeled axes and a legend).

ii. How long do we need to make students wait so that at least one student is still working in the steady state?

Solution:

- As α increases, $x_W[n]$ decays slower.
- For every $\alpha < 0.9$, numerical simulation should still show $x_W[n] \to 0$ eventually.
- Only for $\alpha = 0.9$ does the steady state have any students working (in fact, all students are working who started out working).
- See the solution notebook for an example of a correct implementation and plots.

(f) Instead of an app that makes students wait to use their phones, students to download a different app that reminds them of the existance of work to do while they are on their phone. Note that this app only activates after a student has already started using EdStem, Instagram, or Netflix.

We model the "pushiness" of this app by a parameter γ, so that the transition probability for staying on an app decreases by 0.1γ, while the probability of returning to work increases by 0.1γ.

For example, if $\gamma = 1$, the EdStem column becomes $\begin{bmatrix}0 & 0 & 0.9 & 0.1\end{bmatrix}^\top$.

Let $\mathbf{A}(\gamma) : [0, 7] \to \mathbb{R}^{4\times 4}$ be the transition matrix accounting for this app.

i. Derive an expression for $\mathbf{A}(\gamma)$ (i.e. write out the entries in terms of γ and the original entries of $\mathbf{A}$). Is this still a conservative system?

ii. In the python notebook, find approximately the minimum γ such that, for the initial state $\mathbf{x}[0] = [110, 60, 10, 140]^\top$, at least half of the students who started out working are still working in the steady state. You may round your answer to the nearest integer. For this γ, plot $x_N[n], x_I[n], x_E[n], x_W[n]$ versus n on one figure with labeled axes and a legend.

Solution: For making the students wait γ minutes $\mathbf{A}(\gamma)$ is given by:

$$\mathbf{A}(\gamma) = \begin{bmatrix} 0.8 - \frac{\gamma}{10} & 0.3 & 0 & 0.4 \\ 0.2 & 0.7 - \frac{\gamma}{10} & 0 & 0.1 \\ 0 & 0 & 1 - \frac{\gamma}{10} & 0.4 \\ \frac{\gamma}{10} & \frac{\gamma}{10} & \frac{\gamma}{10} & 0.1 \end{bmatrix}$$

This is still a conservative system as each column sums to 1. You should find that $\gamma = 3$ is the minimum whole number of minutes that we need to block students so that at least half of the students who started out working are still working in the steady state. At 3 minutes of blocking, 80 students are still working in the steady state. Any number of minutes between 2.5 and 3 is an acceptable answer. See the solution notebook for an example of a correct simulation and plots.

6. Homework Reflection

Doing homework is an **investment in your education**, and the course staff wants to hear about your experience with this problem set! After finishing the rest of the homework, fill out these brief reflections on what you got out of your time. We'll read this and take your answers into account as we write future assignments, in this semester and future ones.

Note that we are genuinely curious to read what you have to write. We expect brief but thoughtful responses. **Hence, this part will be weighed significantly in your homework completion grade.**

(a) Did you work on this homework alone or in a group? Did this work well for you? We encourage you to attempt each problem on your own first, and to work with a group when you're stuck.

Solution:

I worked alone because my study group was out of town.

I prefer to work alone because a group can be very distracting for me.

I worked with my friends Alice and Bob. It's nice to be able to bounce ideas off one another and fill in each other's gaps. Explaining my ideas to someone else is hard, but it really makes sure I understand them well!

I met some new friends Carol and Danny at homework party, and we were able to finish most of the homework together with occasional help from a tutor. I reviewed my answers on my own to make sure I understood everything.

(b) What resources did you use to solve the homework problems? How did you use these resources, and how did they help you? Feel free to mention LLM tools, if you've used any.

Solution:

I didn't remember some of the definitions I learned in linear algebra, so I consulted my old notes and reviewed concepts as I needed to while solving the problems

I went to office hours to work on the homework, and whenever I got stuck, I asked one of the tutors to provide some hints or review concepts I didn't remember.

I tried all the problems on my own, and then asked an LLM to critique my approach. I got 90% of the way there on my own, but this interaction enriched my understanding of the material!

I reviewed my lecture notes before starting the homework; and whenever I got stuck, I scoured the discussion worksheets for similar problems and used the solutions to guide my thought process.

I asked some questions on Ed whenever I was confused so that I could get the most fitting answer to my question, in the context of EE66. I also answered some questions because this is a good way to practice the stuff I know well.

I couldn't even understand what problem 2 was asking, so I asked an AI tool to review all the relevant prerequisite knowledge and explain the problem statement in a different way before trying to tackle it on my own.

(c) What was the most fun / interesting problem on this homework? What was the least fun / un-interesting problem?

Solution:

I found question 3 very intriguing because this is a totally different perspective on DTFS to what we learned in lecture. Question 2 was pretty boring and mechanical, because we were just practicing the synthesis equation, but I suppose it is important to be able to do these more-basic things too.

(d) Did you feel prepared to tackle these problems? Were there any problems that seemed completely disjoint from what you've seen in class? Was there anything covered in class that you weren't able practice in this assignment?

Solution: I did feel prepared! But, question 5 seemed to me to be out of place because it's been two weeks since we covered least squares. We talked about SVD and its application for PCA in class and discussion this week, and I didn't really understand any of it, so it would have been nice to have some problems which gave me an opportunity to practice those concepts.

(e) Is there anything else you would like to share?

Solution:

Johnny, who is a tutor, was SUPER DUPER helpful in office hours. He explained stuff in such a way that I still felt like I was discovering the answer to the problems by myself, which was awesome.

I'm feeling very overwhelmed by this class and the upcoming midterm is really stressing me out. These homeworks are really hard and I wish there were more resources available, which I could use to understand them deeply.

Temporary page!

LaTeX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because LaTeX now knows how many pages to expect for this document.